TECHNICAL ARTIST PRACTICAL TEST

CANDIDATE PRACTICAL TEST
TIME LIMIT: 2 DAYS MAX

Project: **TopDown Tank Battleground** — Shaders, VFX, Lighting & Performance

Target Role: Technical Artist (Unity)	**Allowed Time: 2 Days Max** (Approx. 6 – 8 hours of work)
Starting Baseline: Unity 2021.3 Built-in RP (Flat Prototype)	**Target Pipeline:** Universal Render Pipeline (URP) on modern LTS
Core Focus: Shaders, Lighting, Post-Processing, VFX, Profiling	**Deliverables:** Source Repo + Playable Build ZIP + Optimization Notes

Candidate Mission Brief: You are handed an early prototype of a top-down arcade tank game (*TopDownTank-main*). Mechanically the game works, but visually it is completely unpolished: flat greybox prototype cubes, default uncalibrated lighting, no post-processing volume, zero stylized shaders, and bare particle explosions that lack impact and cause fill-rate spikes. Your task over the next **2 days** is to transform this greybox prototype into a vibrant, stylized, arcade combat arena. You will upgrade to URP, build custom shaders for tanks and arena geometry, establish mood through lighting and post-processing, overhaul weapon and impact VFX, and optimize the rendering pipeline for rock-solid framerates.

1. Current Visual State & Technical Art Gaps

An inspection of the assets and rendering configuration in TopDownTank-main highlights several key areas needing your expertise:

Visual Area	Current State in Project	Technical Art Problem & Goal	Priority
Render Pipeline	Legacy Built-in RP with standard materials in Assets/.	Missing modern Universal Render Pipeline (URP) features: SRP Batcher, volume-based post-processing, and flexible Shader Graph workflows.	HIGH
Vehicle Shading	Generic standard diffuse materials on tank hull and turret.	Lacks stylized character. Needs custom shaders (e.g. Toon/Cel shading, rim lighting, team color tinting, and damage flash feedback).	HIGH
Environment Art	Flat grey prototype cubes (Ciathyza/Gridbox) and floor.	Arena looks like an untextured test map. Needs stylized surface shading (tactical grid, hex outlines, or stylized textures) and arena framing.	MEDIUM
Weapon & Impact VFX	Basic projectile sphere; single generic explosion particle.	No muzzle flash, no shell tracer/trail, no dust kick-up when driving, and impacts lack energy, punch, and visual hierarchy.	HIGH
Lighting & Mood	1 default directional light; no shadows or color tuning.	Scene feels washed out and flat. Needs atmospheric directional lighting, ambient contrast, and a tuned Post-Processing Volume.	MEDIUM
Rendering Budget	Unbatched materials, potential particle overdraw spikes.	Materials not configured for GPU Instancing or SRP Batcher; unoptimized particle blending can cause frame rate drops on low-end hardware.	HIGH

2. Candidate Tasks & Requirements

Below are the primary technical art tasks to complete over the 2-day period. Balance visual flair with clean performance.

TASK 1: URP Pipeline Upgrade, Lighting & Post-Processing PIPELINE & MOOD

Goal: Upgrade the project to Universal Render Pipeline (URP) on modern Unity LTS and establish a cohesive visual atmosphere.

• **URP Configuration:** Set up a clean URP Asset with balanced settings (HDR, Anti-Aliasing, appropriate Shadow Cascades & Distance).

• **Material Conversion:** Convert existing materials cleanly to URP (ensure zero broken magenta shaders in the scene).

• **Lighting Setup:**

- Directional Key Light with warm sunlight/moonlight and clean contact shadows.
- Colored ambient/rim fill to give 3D forms depth from the top-down camera angle.

• **Global Post-Processing Volume:**

- *Bloom:* Tuned threshold for energetic glowing projectiles, lasers, and explosions without blinding blowouts.
- *Tonemapping & Color Adjustments:* ACES or Neutral tonemapper with vibrant, rich contrast.
- *Vignette & Ambient Occlusion:* Subtle vignette to frame the arena and SSAO to ground obstacles and tanks on the floor.

TASK 2: Custom Shaders (Stylized Tanks & Arena Floor) SHADER CREATION

Goal: Author custom shaders (using Shader Graph or HLSL) that give the game an intentional, stylized visual identity.

• **1. Stylized Vehicle Shader (Player & Enemy Tanks):**

- *Toon / Stylized Shading:* Stepped diffuse lighting or stylized ramp texture for clean, non-photorealistic look.
- *Dynamic Team Color Tint:* Expose properties to tint player (e.g. vibrant green/cyan) and enemy (e.g. orange/crimson).
- *Fresnel / Rim Lighting:* Subtle rim glow to make vehicles pop against the ground plane from top-down angles.
- *Hit-Flash / Damage Feedback:* A shader parameter (e.g. _HitFlashAmount) that smoothly pulses white or red when taking damage.

• **2. Stylized Arena Ground / Grid Shader:**

- Procedural tactical grid, honeycomb/hex pattern, or stylized tiled surface that replaces flat prototype cubes.
- World-space UV mapping or triplanar projection to eliminate texture stretching across arena barriers.

• *(Bonus):* Dissolve or burn-away effect when tanks or obstacles are destroyed.

Shader Architecture & Parameter Matrix

Shader Target	Visual Techniques Used	Exposed Material Properties
Stylized Tank	Toon Ramp, Fresnel Rim Light, Hit Flash, Specular Cutoff	_BaseColor, _RampThreshold, _RimColor, _HitFlash
Arena Floor & Grid	World-Space UVs, Procedural Grid Lines, Subtle Edge Fade	_GridSize, _GridLineWidth, _GridColor, _GroundTint
Projectile Energy	Additive/Alpha Glow, Fresnel Core, UV Pulse Animation	_EnergyColor, _GlowIntensity, _PulseSpeed

2. Candidate Tasks (Continued)

TASK 3: Complete Visual Effects (VFX) Overhaul VFX & PARTICLES

Goal: Build a responsive, readable visual effects suite for shooting, projectile flight, impacts, and vehicle motion.
• **1. Cannon Muzzle Flash:** Bright instant flash, expanding smoke puff, and shell ejection spark at the barrel tip.
• **2. Projectile Flight:** Energetic projectile model with a ribbon/trail renderer and emissive HDR glow that leaves a glowing wake.
• **3. Multi-Layered Impact Explosion:**
- Core flash & shockwave ring expanding along the ground plane.
- Secondary flying embers / sparks with gravity and bounce.
- Lingering dissipation smoke puff (with soft particles to prevent harsh clipping into the floor).
• **4. Driving Dust / Tread VFX:** Particle dust kicks up from the tracks when the tank turns or accelerates.
• **Performance Note:** Control particle lifetime and maximum counts to avoid particle overdraw spikes.

TASK 4: Visual Juice, Motion & Screen Feedback JUICE & MOTION

Goal: Bring the gameplay alive through animation principles, secondary motion, and dynamic visual feedback.
• **Turret Barrel Kickback:** Procedural kickback or animation on the cannon barrel when firing (quick snap backward, smooth spring forward).
• **Screen Shake Integration:** Wire Cinemachine Impulse to trigger satisfying screen shake on cannon shots and heavy explosions.
• **Stylized Aiming Reticle:** Polished in-world reticle with smooth lead indicator or target acquisition animation.
• **Damage Feedback:** Red hit-flash vignette or screen glitch when the player takes damage.

TASK 5: Tech Art Profiling & Optimization Report PERFORMANCE BUDGET

Goal: Ensure the enhanced visuals run smoothly at 60+ FPS without hitching, and document performance budgets.
• **SRP Batcher & GPU Instancing:** Verify that all custom shaders are compatible with the SRP Batcher and GPU Instancing.
• **Overdraw Audit:** Check the Scene View Overdraw mode to ensure particle systems do not create opaque fill-rate bottlenecks.
• **Texture Compression & Atlasing:** Configure proper texture formats (e.g. BC7/ASTC) and power-of-two resolutions.
• **Written Tech Art Report (TECH_ART_OPTIMIZATION.md):**
Provide a concise 1-page summary covering: Draw call count before/after, shader instruction count, lighting/shadow budget, and profile screenshots from the Unity Frame Debugger.

TASK 6: Playable Standalone Build & Beauty Renders FINAL DELIVERABLE

Goal: Export a working Windows 64-bit standalone build demonstrating the complete visual overhaul.
• Export a standalone executable (.exe) and verify lighting, post-processing, and shaders compile cleanly outside the editor.
• Include 3 to 4 high-resolution beauty screenshots (Before vs. After) or a short gameplay GIF in the repository.

3. Scoring Rubric & Submission Checklist

Submissions are evaluated on a 100-point rubric focused on visual quality, technical execution, and rendering performance.

Category	Points	Evaluation Criteria
1. Pipeline & Lighting	20 pts	Clean URP setup with zero magenta shaders. Cohesive lighting mood and well-balanced post-processing (Bloom, Color Grading, AO).
2. Custom Shaders	25 pts	Stylized tank shader (toon ramp, rim glow, hit-flash pulse) + arena floor shader. Clean Shader Graph / HLSL with exposed properties.
3. VFX & Particle Feel	20 pts	Impactful cannon fire, glowing shell trail, multi-layered explosion (shockwave, sparks, smoke), and track dust. Good visual hierarchy.
4. Motion & Juice	15 pts	Barrel kickback recoil, screen shake integration, stylized reticle, and clear damage feedback.
5. Tech Art Optimization	20 pts	SRP Batcher compatibility, minimal particle overdraw, proper texture compression, and concise TECH_ART_OPTIMIZATION.md report.
Total Score	**100 pts**	**Passing score: 75+ points.** ***(Bonus: +5 points for tank tread decal marks, dissolve effect, or animated track UVs).***

Submission Procedure & Checklist (2-Day Window)

Deliverable	Details	Status
1. Git Repository	GitHub/GitLab repository link (exclude Library/, Temp/, and Logs/).	**[ ] Done**
2. Standalone Build	Playable Windows 64-bit build (.exe) zipped and shared via Google Drive or GitHub release.	**[ ] Done**
3. Tech Art Report	A short TECH_ART_OPTIMIZATION.md detailing draw calls, shader complexity, and Frame Debugger notes.	**[ ] Done**
4. Beauty Screenshots	3 to 4 high-res screenshots showing off the upgraded lighting, shaders, and combat VFX.	**[ ] Done**

Practical FAQ for Candidates

Q: May I use Shader Graph or must I write code in HLSL?
A: Either is completely acceptable. Unity Shader Graph is encouraged for fast iteration and readability. If you write HLSL directly, ensure it is cleanly structured with URP Core support.

Q: Can I use third-party particle textures or meshes?
A: Yes, you may use free textures, smoke sprites, or noise textures for your VFX. However, all shaders, materials, and particle system configurations must be authored by you.

Q: What is the most important evaluation criteria?
A: Visual cohesion and game feel! We want to see how you turn a flat, sterile prototype into a punchy, stylized experience while keeping rendering performance under control.